

R Basics

Lecture 1

Dr. Eric Friedlander

Before we start...

- Make sure you have done the following:
 - Downloaded and installed R
 - Downloaded and installed Positron (or R Studio)
 - Created a Gemini Pro AI account (free for students)

Introduction to R

What is R?

- R is a statistical programming language
 - Inspired by the S programming language developed at Bell Labs in the 1970s
 - Created by Ross Ihaka and Robert Gentleman at the University of Auckland, New Zealand in the early 1990s for teaching statistics
 - Now one of the most popular programming languages for data science and statistics
- R is free and open source
 - Anyone can download and use R for free
 - Anyone can contribute to the development of R and its packages

R is not the same as R Studio, R Markdown, or Positron

- R is the programming language
- R Studio (now Positron) is an integrated development environment (IDE) for R
 - Provides a user-friendly interface for writing and running R code

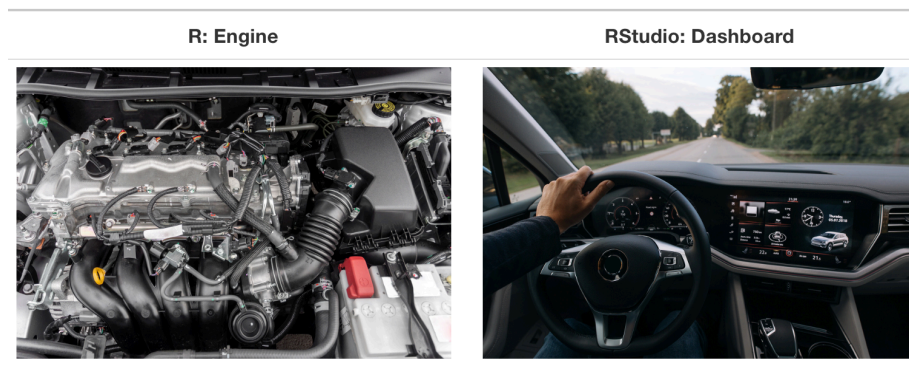


Figure 1: Source: [Statistical Inference via Data Science](#)

Working with R

Writing Scripts in R

- Let's open up a script and run some R code!
 - Running R from the console
 - Running R from a script file

R Packages

- R has a rich ecosystem of packages that extend its functionality
 - You can install packages using `install.packages("package_name")` (only do this once)
 - Load packages into your R session using `library(package_name)` (do this every time you start a new R session)
- An important package for this class:
 - **tidyverse**: A collection of packages for data manipulation, visualization, and modeling
 - * **ggplot2**: A package for creating graphics using the Grammar of Graphics
 - * **dplyr**: A package for data manipulation
- Let's install and load **tidyverse**

Palmer Penguins Dataset

- Install the **palmerpenguins** package
- Contains dataset on the famous Palmer Penguins: penguin species, island in Palmer Archipelago, size (flipper length, body mass, bill dimensions), and sex.
- Explore more with `?penguins`

Important Vocabulary

- **Data frame:** A table-like structure in R that holds data (**penguins**)
 - In the **tidyverse** data frames are referred to as **tibbles**
- **Variable:** is a quantity, quality, or property that you can measure (e.g. **species**, **bill_length_mm**, **island**)
- **Observation** or **Data point:** a single object that has been measured (e.g. one of the penguins in the data set)
- **Value:** the entry for a particular variable and observation (e.g. for the first penguin, the value of **species** is “Adelie”, the value of **bill_length_mm** is 39.1)
- **Tabular Data:** data organized in rows and columns (like a spreadsheet or data frame)
- **Tidy Data:** tabular data where each observation is a row, and each type of observational unit forms a table

Taking a quick look at penguins

- **glimpse** gives a quick overview of the data frame
- **head** shows the first few rows of the data frame
- **tail** shows the last few rows of the data frame

Wrap up

Complete the following before next class

1. Create a GitHub account if you don't have one
2. Read the [syllabus](#)
3. Make sure you can log in to RStudio and reserve a container for STA 313: cmgr.oit.duke.edu/containers
4. Complete the *Getting to know you* survey on Canvas – to be posted
5. Complete the readings